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Carbon capture and storage at Malaysia power plants: Evaluation of its feasibility and requisite enablers

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ABSTRACT

Rapidly growing GHG emissions are significantly altering the world climate, necessitating a global decarbonization efforts to avert irreversible catastrophic failure of the world ecosystem. Malaysia is pledging to achieve carbon neutrality by 2050 and has subsequently identified Carbon Capture and Storage (CCS) as one of the solutions for carbon removal from its energy sector by 2050. However, the implementation of CCS presents formidable challenges, encompassing both technical and commercial complexities, especially for a nation like Malaysia, which has limited experience in large-scale CCS deployment. This study evaluates the feasibility of CCS implementation in Malaysia's fossil fuel power plants by conducting a comprehensive techno-economic assessment, taking into considerations the prevailing energy landscape of the country. This assessment is very crucial to demonstrate the feasibility of this plan within the timeframe, considering the limited research have been done to assess a national-scale implementation, as well as to provide the important insights for future policy implication to promote CCS adoption. The CCS scenarios are assessed through an analytical approach, and comparative analyses are conducted between alternative paths. The result shows that a large-scale implementation of CCS at Malaysia's fossil fuel power plant to achieve net-zero emission by 2050 are feasible given a strong enforcement on emission policy and establishment of CCS legal framework. CCS can potentially reduce 60% of Malaysia's energy sector total emissions, and a solar-supported CCS helps to reduce it further. Levelized Cost of Electricity analysis reveals that the implementation of CCS leads to only a minimal increase between pre- and post-CCS LCOE when carbon price is considered. A SWOT-PESTLE analysis was also conducted to identify gaps in regulatory framework, financial and social aspects. It can be concluded that CCS is a techno-economically viable solution, however Malaysia's readiness for CCS implementation is still at a very novice stage, and aggressive actions are needed to enable a successful full-scale implementation.

1. Introduction and overview of carbon capture and storage in malaysian energy sector

Increasing GHG emission and its impact to climate change has been recognized by global leaders since the Paris Agreement at the UN Climate Change Conference (COP21) in Paris (2015). Gradually, member-countries begun to set target of emission reduction and making pledges in the subsequent COP. In July 2021, 103 countries including Malaysia renewed their pledges on Nationally Determined Contribution (NDC) of emission reduction at the UN Climate Change Conference 2021 (COP26) to curb global warming and its devastating impact. All the member countries have agreed to pledge financial commitments and action plans to achieve greenhouse gas emission (GHG) reduction targets, forming the Glasgow Climate Pact which has now superseded the Paris Agreement.

1.1. Carbon capture and storage: A decarbonization tool

Carbon capture and storage refers to a chain of processes where CO_2 are captured from an emitting source such as power plants, industrial facilities, oil and gas processing facilities, and then transported and injected into a suitable underground formation for permanent storage as shown in Fig. 1 (Global CCS Institute). It is a technology that is used to minimize the amount of emitted carbon to the atmosphere as a result of processes in these industries. There are three main components of CCS i.e., CO_2 capture, CO_2 transport and CO_2 storage. Each of the component and their respective technologies are tabulated in Table 1. CCS is first used in the Sleipner gas processing facility offshore Norway in 1996 in

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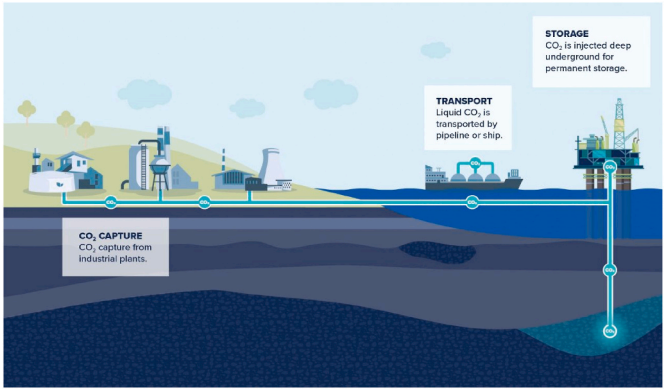


Fig. 1. Carbon capture and storage (global CCS Institute).

Table 1
CCS components and technology.

Component	CO_2 Capture	CO_2 Transport	CO_2 Storage
Technology/ Options	• Post combustion capture • Pre combustion capture • Oxy-fuel combustion capture • Direct air capture	• Via CO_2 Pipeline • By Ships/ Tankers • By Trucks	• Into Saline aquifer • In Sedimentary basin • Into Depleted oil/ gas field

which CO_2 are captured from an offshore gas processing platform and then compressed and reinjected into a saline aquifer above the gas reservoir. Currently there are 29 operating CCS facilities in the world which are implemented across multiple industries (Steyn et al., 2023).

Theoretically, a CCS scheme can capture up to 90% carbon from the total emission of the processes (Kazemifar, 2022). However, there is also a downside to CCS which manifest in a form of energy penalty. CCS processes need a minimum amount of energy to operate and will draw power from the host process. If CCS is implemented in a fossil fuel power plant, a considerable amount of the power generated by the plant will be diverted to the CCS module for its consumptions, which will lead to the reduction of the total electricity output to the grid. It is estimated that implementation of CCS will effectively reduce the power plant efficiency by 14–20% due to this energy penalty (Kazemifar, 2022). Despite this downside, CCS remains the most proven technology to decarbonize industrial emissions.

1.2. Net-zero 2050 in the 12th malaysian plan

As a member country, Malaysia subsequently announced a net-zero target by year 2050 in the RMK-12 in September 2021. A joint study between BCG, WWF and Government of Malaysia reported that since 2004, Malaysia has become a net positive emission country due to the advancement and development (WWF Malaysia and Boston Consulting Group, 2021; Abd Rahman et al., 2022; Alasinrin Babatunde et al., 2018). To achieve this, a set of 10 priorities was established in the Net-Zero Pathways for Malaysia which includes decarbonization of the energy sector as the second priority. Fig. 2 shows the breakdown of Malaysia's emission by source. With 52% of Malaysia total carbon emission contributed by the energy generation in 2016, Carbon Capture and Storage (CCS) at fossil fuel power plants has been identified as one of the green technology solutions to reduce carbon emission contribution from this sector (WWF Malaysia and Boston Consulting Group, 2021). To achieve significant GHG emission reduction and contribute to energy sector decarbonization, CCS will have to be implemented to all fossil fuel power plants in the country. Unfortunately, for a deployment

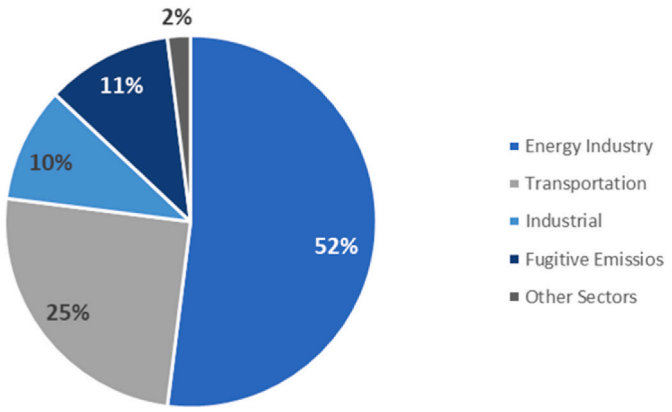


Fig. 2. Malaysia emissions percentage by source (WWF Malaysia and Boston Consulting Group, 2021).

at this scale, the technical feasibility, commercial viability, market fit, and operational feasibility has yet to be proven, thus leaving uncertainty whether the reduction target from energy industry can be met at all.

1.3. Malaysia energy giant, TNB & PETRONAS announces net-zero 2050

In response to this, Malaysia's giant energy corporations i.e., TNB and PETRONAS joined the effort and announces their Net-Zero 2050 target on 26th August 2021 and 1st November 2022 respectively. TNB has pledged around RM20 billion in CAPEX to work on emission reduction from its operation which includes decarbonization of its current power plant through carbon removal technology, expansion of its renewable energy portfolio and development of electric vehicle technology ("TNB aims to achieve net, 2022). While PETRONAS has pledged 20 percent of its CAPEX from 2022 to 2026 to the concentrated effort to decarbonize its operation. This includes reduction of methane gas release trough venting and flaring at its facilities as well as expansion into clean energy solutions (Nizam, 2022). On 19th August 2022, PETRONAS and TNB announced their Join Forces to Advance Malaysia's Carbon Neutral Aspirations and signed an MoU to conduct a joint feasibility study on the application of CCS solutions at a pilot plant ("TNB and Petronas join forces to, 2022). Though assessing the implementation of CCS at one specific plant is a step forward for the technology development, the bottom-line still lies on whether a wide scale implementation of CCS at power plants will work in the Malaysia energy landscape.

1.4. Problem statement

Malaysia's Net Zero 2050 Pathways has included carbon removal technology i.e., CCS as one of the key solutions to achieve the carbon neutrality target in time, without which, will push the achievement timeline to 2055 (WWF Malaysia and Boston Consulting Group, 2021). Unfortunately, as the technology is still evolving, there are still uncertainties and unknowns with regards to techno-commercial viability of a wide-scale CCS implementation. Therefore, there is a need for an evaluation that will provide the essential interdisciplinary assessment of wide-scale implementation of CCS and its feasibility in Malaysia and whether the reduction target from energy industry can be met at all. The following sections describes the issues that needs to be addressed.

1.4.1. Lack of global success in power plant CCS implementation

Although CCS is not a new technology and has been around since 1972 (Staib et al., 2022), the specific implementation at power generation facilities are limited. Compared to CCS implementation at products processing plants, the power plant CCS is clearly lacking in experience and lesson learnt, and the information gathered so far is insufficient to

encourage replicability or recommend best practices. Global CCS Institute in their Global Status of CCS 2022 reported that there are 30 CCS facilities currently in operation around the world, and out of that number only one CCS project is implemented at a power generation facility as shown in Table 2. As of 2022, there is only one power plant CCS in operation i.e., Boundary Dam while another has been suspended i.e., Petra Nova (Steyn et al., 2023).

A total of 32 projects are trailing in the development stage however many have yet to pass the projects hurdle to make an investment decision. This trend implies that the technical feasibility, commercial viability, market fit, and operational feasibility has yet to be proven, thus leaving uncertainty whether a successful implementation can be achieved. It can also be concluded from literature review that most study and research regarding are concentrating of the United States, European as well as China localities. The study of CCS viability in Malaysia is still limited to the small-scale implementation in offshore producing gas field and lacking exploration in the application to the fossil fuel power plant. The body of knowledge is not mature enough to propel the move of a wide-scale deployment.

1.4.2. Limited CCS operational experience in Malaysia

According to IRENA country report, 96% of Malaysia's total energy supply comes from fossil fuel power plants and this contributes to a total of 133 $MtCO_2eq$ emission in 2019 (IRENA, 2022). Currently in Peninsular Malaysia there are 27 coal and gas power plants are in operation with a total installed capacity of 1440 MW, compared to renewables 327 MW (IRENA, 2022). This shows a clear dominance of fossil energy and for the country to completely phaseout the fossil fuel power generation by 2050 is too ambitious. To achieve significant GHG emission reduction and contribute to energy sector decarbonization, CCS will have to be implemented in a wide scale across the country. This will be a major challenge for Malaysia to take on the CCS as the country itself has never embark on any CCS project before. The most advance CCS attempt is Malaysia is the Kasawari CCS project, a gas field production project offshore Sarawak, which is still facing techno-commercial challenges to be monetized.

The journey of CCS in Malaysia started in 2009 when Malaysia officially pledged a 40 % reduction in emission by 2020 based on 2005 level during the COP15 2009 in Copenhagen. Since then, there have been efforts from the government as well as the private sector especially in capability building. To date, the first CCS project is planned to be implemented in the oil and gas industry, the Kasawari CCS is currently at development stage. Recently, Malaysia energy corporation Tenaga National Berhad and PETRONAS has started their joint feasibility study on power plant CCS. Many studies have been conducted on CCS deployment in Malaysia however only a few discusses the feasibility of a full chain CCS implementation. In Malaysia, CCS study was first carried out in 2010 to evaluate a full-chain pilot CCS project in Jana Manjung coal-fired power plant. The study concluded that current technology is sufficient to support the implementation of CCS in Malaysia. Subsequently, the study also highlighted that the absence of adequate funding is the major barrier.

Research in 2015 studied the potential of CO_2 reduction via CCS as well as the cost of CCS however this study is an assessment across several emission industry i.e., energy, oil and gas and industrial sector in Malaysia (Rahman, 2015) (Rahman, 2015). The study also reported that the cost of energy produced by fossil fuel plant with CCS is equal to the cost of energy from renewable resources. In another study by the same researchers, they propose four sources of revenue for Malaysian CCS project business model: carbon trading, carbon tax, EOR and algae production (Rahman, 2015). The rest of the literature are mainly assessing a particular segment of CCS for example Zulaikha et al. (2013) conducted a storage study in which few potential sites in the Malay basin were identified. Despite the early start, the progress of CCS is still not matching with the requirements to meet net-zero ambition 2050. Although many issues inhibiting implementation of CCS were already identified early on, little has been done to address them. To date, there is still absence of regulatory frameworks and policies which is crucial in supporting the much-needed rapid development of CCS in Malaysia.

1.4.3. Complexity of integrating CCS into the existing energy landscape

Power plant CCS involves integrating the carbon capture module into the power plant combustion system and subsequently diverting the captured CO_2 into a compression unit before transporting and sequestrating it into an underground carbon storage. There are currently only two power plant CCS project that has advanced to operation stage i.e. The Boundary Dam CCS and Petra Nova CCS. Due to the low performance of the capture technology, coupled with commercial challenges, the implementation of CCS in a power plant has track record of failure. The major problems of power plant CCS are the high cost of retrofitting an existing plant and the low CO_2 capture rate and the energy penalty of carbon capture. In theory the CCS module can capture up to 90% CO_2, however historically a capture rate as low as 50% has been recorded (Robertson and Mousavian, 2022). Both projects economic models are based on the sales of CO_2 for enhanced oil recovery (EOR) purposes. This has been debated as the CO_2 are injected to produce oil which will eventually produces more CO_2. As for the energy penalty, an average of 14% reduction in power plant efficiency has been reported with the implementation of CCS. Despite the issues, there have been an increase of power plants CCS project globally as industry players see CCS as a 'license' for them to continue to operate on fossil fuel while minimizing the environmental impact (Steyn et al., 2023). Global CCS Institute reported 32 power plant CCS projects are currently in development stage globally (Staib et al., 2022).

The total of capacity of global CCS project has seen an increase from 75 Mtpa to 111 Mtpa from 2020 to 2021. However, despite this encouraging increase, only 2.8 % of this capacity are currently under construction. This slow trend is caused by the complexity faced in integrating CCS into an already established energy landscape. Apart from technological issues like high uncertainty in capture technology and the high uncertainty of the reliability of CO_2 storage site, CCS also faced with commercial and policy issues like inadequate financing incentives, lack of commitment from energy players as well as low public acceptance and political support (Sara et al., 2015). Implementation of CCS in Malaysian power plant will be significantly changing the energy landscape in Malaysia. The current readiness for this shift needs to be examined.

To address the problems defined, this project aims to conduct a multi-disciplinary evaluation on the viability of CCS implementation at Peninsular Malaysia fossil fuel power plants and integrate the information to form the overall techno-economic do-ability of CCS in Malaysia. It will be evaluating the feasibility from technical, commercial and policy point of view. This work aims to achieve a) To quantify the potential carbon emission reduction from implementation of carbon capture and storage at TNB power plants in Peninsular Malaysia i.e., gas and coal as well as other benefits, b) To estimate cost for capture,

Table 2
Global CCS facilities status (Steyn et al., 2023).

	Operation	Construction	Development	Suspended
Number or Facilities	**30**	**11**	**153**	**2**
Power Generation	1	1	32	1
Natural Gas Processing	13	3	12	1
Cement Production	0	2	3	0
Iron & Steel Production	1	0	0	0
Waste to Energy	0	1	4	0
Direct Air Capture	1	2	2	0
Hydrogen Production	2	0	19	0
Chemical Production	1	0	7	0
Fertilizer Production	4	0	4	0
Others	7	2	70	0

compression, transport, injection, and storage of CO_2 and potential offset from renewable energy resources for carbon capture energy cost, c) To analyse the potential barriers of implementation based on lessons learnt from prior projects around the world and d) To investigate the key enablers for implementation of carbon capture in Malaysia and the design consideration for future power plants to successfully integrate carbon capture and storage technology.

1.4.4. The challenges, opportunity and breakthrough of large-scale CCS

Carbon capture at power plants involved the process of complex post-combustion molecules capture which comes with many challenges. Among the challenges and barriers that have been encountered in implementing carbon capture technologies at scale includes technological limitations such as high costs, energy penalties, and the need for improved capture efficiency and storage capabilities. Moreover, regulatory and policy frameworks often lack sufficient incentives or mandates to spur investment and deployment of carbon capture infrastructure. However, amidst these challenges, significant opportunities have emerged, driven by advancements in carbon capture technology and growing recognition of the importance of decarbonizing the energy sector. Recent breakthroughs in materials science, process engineering, and carbon utilization have enhanced the viability and efficiency of carbon capture systems, offering new pathways for cost reduction and improved environmental performance. Additionally, international collaborations and initiatives have fostered knowledge exchange and resource sharing, accelerating progress in carbon capture research and deployment. As nations strive to meet ambitious climate targets, the development and deployment of carbon capture technologies are poised to play a pivotal role in achieving a sustainable and low-carbon future. Malaysia can definitely leverage on these breakthroughs to advance its CCS aspirations to achieve net-zero emission by year 2050.

2. Methodology

The evaluation is divided into five work stages where each work stages are designed to achieve one of the stated project objectives above. The first step taken is to formulate Malaysia's Power Plant CCS concept. This involves the planning of infrastructure layout between the power plants to the carbon storage locations, selection of transportation method, as well as the strategy for managing the substantial scale of nationwide CCS development. The idea of solar-supported CCS scheme is also explored in this stage. The outcome of this work stage is four distinct scenarios that will be used further in the analysis to demonstrate the feasibility of CCS. In the second work stage, the energy generation by sources and emission for each of the four scenarios are forecasted. The energy generation and emission forecast are subsequently feed into the cost analyses in the third work stage. Comparative analyses are then conducted in stage four to identify the best scenario that will give the greatest benefit in terms of emission reduction and overall cost-effectiveness. Finally, a qualitative analysis using a combined SWOT-PESTEL method is conducted to identify the barriers and enablers for power plant CCS in Malaysia. The work steps are illustrated in Fig. 3 and further elaborated in the following sections.

2.1. Development of Malaysia power plant CCS concept

A total of 43 fossil-fuel power plants located in 14 Malaysian states are included in this study. The location of the storage site considered is in the Malay Basin where a few depleted oil and gas reservoir has been identified as the most suitable for CO_2 storage based on a set of evaluation criteria (Junin, 2020). Based on this information, next step taken is to form the CCS infrastructure concept that can be adopted to connect the CO_2 capture facilities at respective power plants to the storage site located offshore Terengganu. A few considerations have been taking into account including:

- The types of infrastructures required and how should they strategically be set up.
- Most cost-effective way to transport CO_2 from capture facilities to the storage site.
- A realistic implementation strategy that can be adopted to minimize risk and increase chance of success.

Leveraging from global experience, two existing power plant CCS projects are reviewed to assess if their concept can be adopted and replicated for Malaysia. The lesson learnt and best practices from these projects also helps to formulate a more robust CCS project. The two existing CCS scheme reviewed are i.e. The Petra Nova CCS in the United States and the Northern Light CCS in Norway. These projects differ in their approaches i.e., Petra Nova operates as a standalone scheme, whereas Northern Light is structured as a clustered development. They

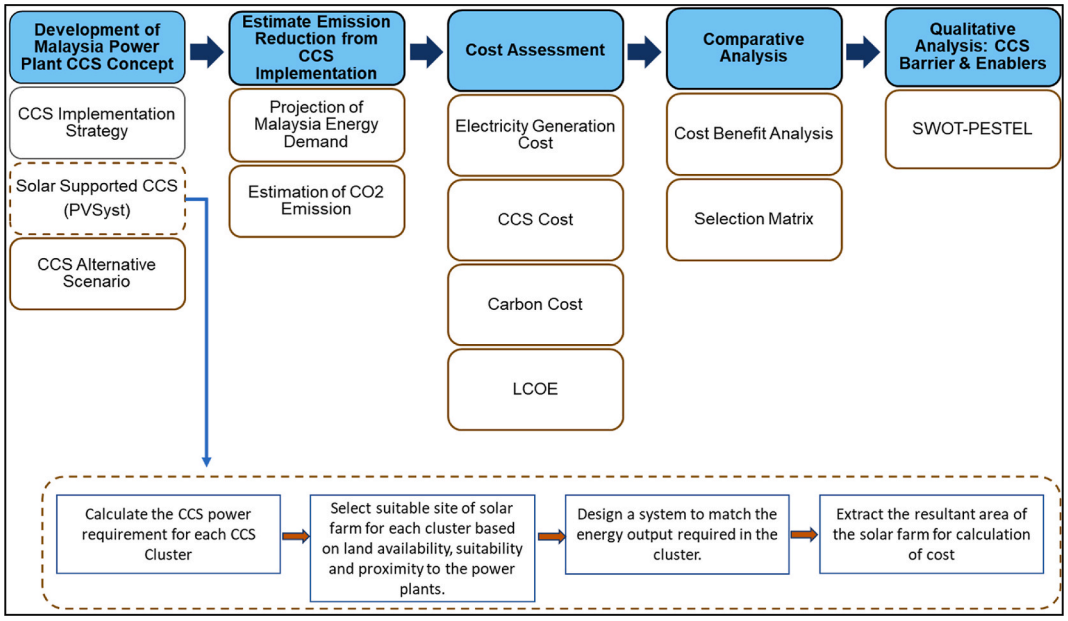


Fig. 3. Workflow and methodological stages.

also employ distinct mechanisms for the transportation of CO_2 i.e., pipeline vs ship as shown in Fig. 4. The attributes of these two CCS are summarized in Table 3. From the information gathered about existing power plant CCS concept, evaluation is conducted based on their strengths and weaknesses within the Malaysian context to determine their relevance and applicability. The outcome of this step shows that the Northern Light concept is the best concept to be adopted in Malaysia.

2.2. CCS implementation strategy

A full-scale CCS deployment in Malaysia constitute a massive undertaking that requires substantial amount of investment and carries inherent risk. It is unrealistic to assume that the construction of nationwide CCS infrastructure can be completed all at once. To increase the chance of successful CCS implementation, the approach to develop the infrastructure must be strategized to minimize risk and maximize the learning curve. Incorporating the lesson learnt and best practices from prior CCS projects (Robertson and Mousavian, 2022), this study adopted the clustered CCS layout and propose the development to be in stages. Power plants will be grouped into one cluster together based on their geographic proximity. This allows for an optimized infrastructure approach where a centralized compression and transportation hub can be shared instead of constructing numerous small-scale facilities to serve individual power plants. This conceptual framework is represented in the Fig. 5. Twelve CCS clusters have been identified along with their implementation timeline, which are elaborated in Section 4.

2.3. Solar -supported CCS versus conventional CCS

One of the downsides of CCS is the additional power requirement that is needed by the CCS module to run its process. This additional power is mainly used in the CO_2 separation process as well as compression of the CO_2 stream for transport. The estimated additional power demand amounts to 11–40% on top of the plant's normal output (Vasudevan et al., 2016). In conventional CCS setup, this extra power requirement is subtracted from the plant's output, resulting in a reduction of electricity supplied to the grid and consequently decreasing the overall plant efficiency. Alternatively, the plant may also consume additional fuel to maintain its normal output levels, but this would inevitably result in increase of emissions.

To address this challenge, this study proposes a third option i.e., the integration of a renewable energy system to meet the power demands of the CCS process. Solar energy is the chosen renewable energy resource considering the substantial solar energy potential in Malaysia, as compared to wind energy, and the flexibility it offers in terms of geographical location compared to hydropower. The proposed development is to link each cluster to a dedicated solar farm tailored to meet the CCS module's specific power requirements. Twelve solar farms are designed to respectively serve each of the CCS cluster.

The sizing of the solar farms is conducted using PVSyst software. Since solar energy is only available during daylight hours, the capacity of the solar system must be designed to sufficiently offset the amount consumed at night. The excess power generated will be injected to the grid, hence levelling the fuel intake back to pre-CCS level. The parameter of interest from the PVSyst simulation is the area required for the solar farm to sufficiently offset CCS power requirement. This parameter will be used to evaluate the feasibility of building mega scale solar farm in terms of area required. The following steps are taken during solar farm sizing in PVSyst which is also depicted in Fig. 3. The simulation results are elaborated in Section 4.

-Step 1: Calculate the CCS power requirement for each CCS cluster.
-Step 2: Select a suitable site of solar farm for each cluster based on land availability, suitability, and proximity to the power plants.
-Step 3: Design a system to match the energy output requires in the clusters.

Table 3
Petra Nova vs Northern Light Concept (Equinor, 2019), (Jenkins, 2015).

Component	Existing Solution	
	Petra Nova CCS	Northern Light CCS
CO_2 Capture Mechanism	Post combustion CO_2 absorption using amine-based solvent followed by stripping of CO_2	Post combustion CO_2 capture potentially using chemical-based method.
CO_2 transport Mechanism	CO_2 is compressed and transported via pipeline to storage or utilization location.	CO_2 is compressed into liquid from and stored temporarily before being transported by CO_2 tanker ships to CO_2 receiving terminal
Onshore/ Offshore facilities location	- Onshore CO_2 capture and compression plant -Onshore injection facilities	-CO_2 capture at powerplants -Onshore CO_2 compression hub -Onshore receiving terminal -Offshore sub-sea injection facilities
Storage Site	CO_2 captured are utilized for enhanced oil recovery project and also stored in underground formation.	CO_2 captured are injected into depleted underground formation in the North Sea.

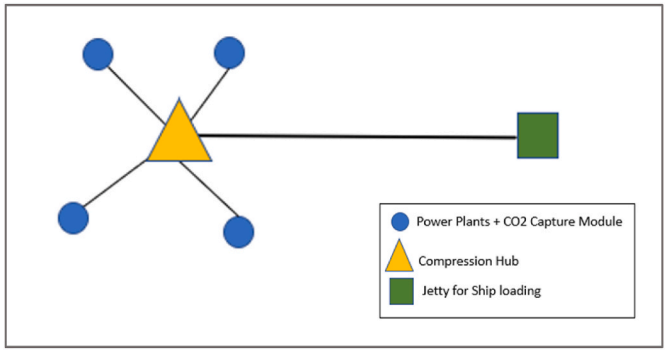


Fig. 5. CCS cluster concept.

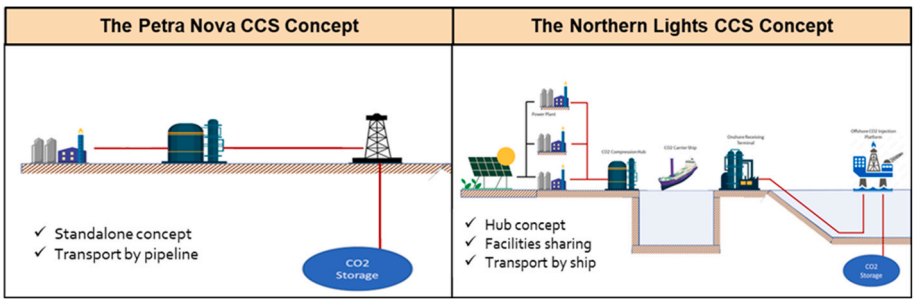


Fig. 4. Different approach and distinct mechanisms for the transportation of CO2.

-Step 4: Extract the resultant area of the solar farm.

2.4. CCS and alternative scenarios

To further evaluate the feasibility of power plant CCS in Malaysia, four scenarios are established. These scenarios serve as a basis for comparing the benefits and costs of CCS-integrated cases against those without CCS integration. Two of these scenarios involve the implementation of CCS, while the remaining two scenarios exclude CCS implementation. The following section details the four scenarios. In the subsequent section of the study, the energy generation forecast, emission forecast, and cost calculation will be conducted for each of the scenario. The 4 scenarios are shown in Fig. 6.

2.4.1. Scenario 1: business as usual (BAU)

This scenario serves as the baseline case upon which the evaluation of the other scenarios will be conducted. In this scenario, no additional measures are taken beyond the current state. It represents the situation in which Malaysia continues to primarily generate its power from fossil fuels without undertaking any emissions-reduction efforts. There is no implementation of CCS, and as a result, CO2 continue to be emitted into the atmosphere. The assumption of energy generation in this scenario is that percentage of fossil fuel contribution at the 2021 level will remain constant until 2050.

2.4.2. Scenario 2: conventional CCS

In this scenario, a full-scale CCS system is installed across all fossil fuel power plants nation-wide. To meet the increased energy demand resulting from CCS implementation, additional fossil fuels are consumed at each respective plant. As a consequence of this extra combustion, the total emissions prior to capture exceed those of the BAU scenario. This scenario represents the conventional CCS design approach, where supplementary fuel is burned to support the CCS function. The development strategy employed for the twelve CCS clusters are gradually brought online one by one, spanning a period of twelve years.

2.4.3. Scenario 3: solar-supported CCS

Scenario 3 shows the option where the CCS implementation are supported by renewable energy to cater for the high energy penalty of post-combustion carbon capture. Instead of relying on burning more fossil fuels, renewable energy in the form of solar power will be utilized. Each CCS cluster will be connected to a dedicated solar farm, designed with ample capacity to satisfy and counterbalance the CCS needs. This will consequently eliminate the additional emissions stemming from CCS implementation, which was the downside of Scenario 2.

2.4.4. Scenario 4: Renewable Energy Growth

This scenario represents the case where the growth of renewable energy is being prioritized over the effort to capture carbon from fossil fuel combustion. This scenario represents the alternative approach to reduce emission from the energy generation sector. Instead of implementing CCS to manage the growing emission from increasing energy demand, solar farms are built to cater for the growing requirement. Under this scenario, it is assumed that there will be no further expansion of fossil fuel power plant generation commencing in 2024. Consequently, emissions are capped at their current levels until 2050.

2.5. Projection of energy generation for four scenarios

Malaysia's electricity generation is forecasted up to year 2050. There will be two cases for the projection. Scenario 1, Scenario 2 and Scenario 3 described above share a same energy generation forecast i.e., Case 1. While Scenario 4 is an alternative profile where fossil fuel generation will be capped at current capacity while ramping up solar energy capacity to support the growing demand, i.e., Case 2. The historical electricity generation data are used to have a more accurate projection of emissions in the country. The historical data are collected from IEA Electricity Information 2022. Data is tabulation in Table 4.

Case 1 electricity generation forecast is adapted from the energy projection study conducted by the Economic Research Institute for ASEAN and East Asia (ERIA) in the Energy Outlook and Energy Savings Potential in East Asia (2020), Malaysia Country Report. This projection considers the growth of governing economics parameter such as GDP, crude oil prices, population, index of industrial production and Malaysia national energy plans hence providing a high confidence forecast. Data is tabulated in Table 5.

The table provides a projection of Malaysia's electricity generation from various sources up to the year 2050. It shows an increase in electricity generation across all sources, with coal, natural gas, and

Table 4
Malaysia historical energy generation by source (IEA).

	Malaysia Historical Electricity Generation, GWh					
	Coal	Oil	Natural gas	Hydro	Solar PV	Biofuels
1990	2932	10555	5540	3989	0	0
1995	3450	8974	26816	6222	0	0
2000	7691	3600	50998	6966	0	0
2005	19991	2203	55287	5191	1	0
2010	42839	3670	70795	6472	0	1002
2015	63474	1739	69962	13924	273	751
2020	85815	953	66222	26217	2338	1022

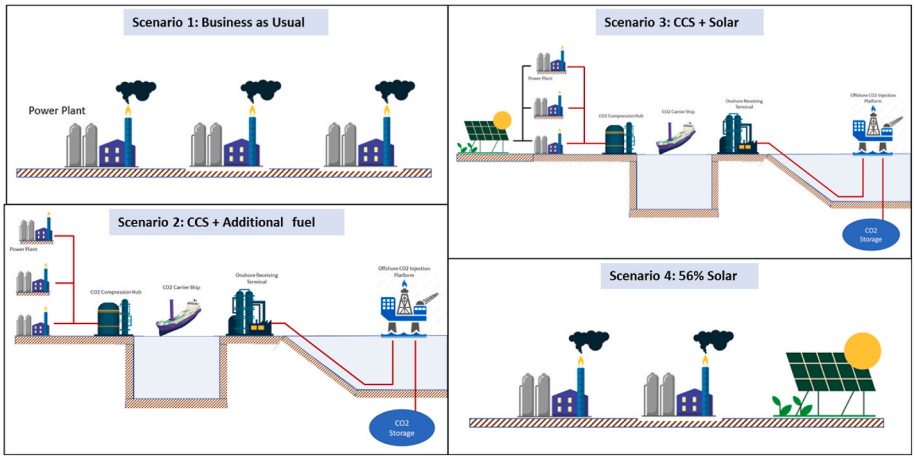


Fig. 6. Scenario 4 renewable energy growth.

Table 5
Malaysia projected energy generation by source (Kimura and Phoumin, 2020).

Year	Malaysia Projected Electricity Generation, GWh				
	Coal	Oil	Natural gas	Hydro	Solar PV + Biofuel
2030	9836	266	11387	3236	487
2040	14470	258	16252	3235	487
2050	18464	259	20745	3236	487

hydroelectric power being the dominant contributors. Specifically, there is a significant growth in electricity generation from coal and natural gas, indicating a continued reliance on fossil fuels for energy production. Additionally, there is a consistent contribution from hydroelectric power, while solar PV and biofuel sources remain relatively constant throughout the projection period. Overall, the data suggests that Malaysia's energy landscape is expected to evolve with a continued emphasis on fossil fuel-based generation alongside sustained growth in renewable energy sources.

Since the projection data are only available for year 2030, 2040 and 2050, the growth is distributed evenly between the data points to get the yearly figures. Data checked was conducted to ensure continuity from historical to projection. For Case 2 electricity generation forecast, the electricity generation from coal, oil and gas will be capped starting from 2023 while solar energy contribution is increased to replace growth of fossil energy. The resultant forecasts are elaborated in Section 4.

2.6. Estimation of CO_2 emission for four scenarios

Each of the scenarios described in Section 3.5 resulted in distinct emission levels depending on the type of energy resources utilized as well as the presence or absence of CCS implementation. There are four cases of emission profile following the four scenarios. The annual emission is calculated based on the projected amount of electricity generation from coal, gas and oil in the year, using published emission factors as tabulated in Table 6. The emission factor estimates the amount of CO_2 emitted per unit of energy generated from each of the fossil fuel. Coal has the highest emission factor due to high carbon content, followed by oil and gas. The emission from coal, gas and oil plant are calculated using equations (1)–(3) respectively.

$$Emission\,from\,coal\,plants = Electricity\,production\,(GWh)\,x\,Heat\,rate\left(\frac{BTU}{GWh}\right) x\,Coal\,Emission\,Factor\left(\frac{kgCO_2}{GWh}\right) \quad (1)$$

$$Emission\,from\,gas\,plants\,\left(kgCO_2\right) = Electricity\,production\,(GWh)\,x\,Gas\,Emission\,Factor\left(\frac{kgCO_2}{GWh}\right) \quad (2)$$

$$Emission\,from\,oil\,plants\left(kgCO_2\right) = Electricity\,production\,(GWh)\,x\,Oil\,Emission\,Factor\left(\frac{kgCO_2}{GWh}\right) \quad (3)$$

The emission forecast for Scenarios 1, 2 and 3 are based on Case 1 energy generation profile while Scenario 4 is based on Case 2 electricity production projection. Although the pre-CCS emission for Scenario 2 and Scenario 3 are the same with Scenario 1, further adjustment is done to Scenario 2 and 3 to reflect the amount of emission captured by CCS. It is estimated that CCS are able to capture 90% of the emitted volume. The emission for Scenario 2 post CCS also includes the emission for additional fuel consumption to support CCS energy requirement. The four resultant emission profiles are elaborated in Section 4.

Table 6
Malaysia emission factors by source (EFDB Editorial Board, 2023).

Fuel	Type	Emission Factors
Coal	Malaysia Bituminous Coal	7083.333 kg CO2/GWh
Gas	Malaysia CCGT Power plant	490,000 kg CO2/GWh
Oil	Diesel Combustion	5500 kg CO2/GWh

2.7. Cost assessment for four scenarios

2.7.1. Electricity generation cost

The electricity generation cost for the four scenarios is estimated according to the electricity resource. Malaysia has six resources of energy i.e., coal, gas, oil, hydro, solar and biogas or biomass. The generation cost varies between the resources type according to the real cost of operation as well as infrastructure needed for electricity generation. For this study, electricity generation cost factor is used to estimate the generation cost on yearly basis, based on the amount of electricity generated from each of the resources. The cost factor represents the capital and operational expenditure spent per unit of energy generated. The electricity production forecast of each scenario is multiplied with these factors to obtain the generation cost of each year. Table 7 summarizes the cost factor used by energy source.

2.7.2. CCS cost

For Scenario 2 and Scenario 3 where CCS is implemented, the cost of CCS will be included. CCS cost is divided into three components i.e., CO_2 capture cost, CO_2 transport cost as well as CO_2 storage cost. Cost component of CCS project includes capture cost, transportation, and storage cost. Due to absence of CCS project within the Southeast Asia Region, the cost estimates are based on the reported cost factors of capture, transport and storage in United States and EU as reported by the Global CCS Institute as tabulated in Table 8. The cost factors represent the capital and operational expenditure spent per tonne of CO_2 captured, transported and stored respectively.

Based on the cost factors above, the CCS cost for coal plants is USD 108.27 per tonnes CO_2 while the CCS cost for gas plants is USD 128.99 per tonnes of CO_2. Since data for oil-fired plant CCS are not available, it is assumed to have the same cost factor as coal-fired plant. The most dominant cost component in CCS is the capture and compression cost because this is the most energy intensive process in the CCS scheme where CO_2 in gas phase is liquefied for transport and injection. The higher cost associated with compressing gas-fired power plant emissions can be attributed to the need for multiple compression stages. These stages are needed to compress the relatively lower concentration of carbon dioxide in the flue gas stream, as compared to coal-fired and oil-fired flue gas. The cost for transport using pipeline is higher than cost of transporting via ship by 72% for transport distance reaching 1500 km (European Technology Platform for Zero Emission Fossil Fuel Power Plants). This explains transportation by ship is more favourable for Malaysia whose power plants are mostly located close to the coastline.

2.7.3. Carbon cost

The effort to decarbonize the energy sector need to be supported by the consideration of the true cost of carbon to businesses, the nation and environment. Without considering the true cost of carbon, there will be

Table 7
Electricity generation cost factor by source (Irena, 2021).

Energy source	Cost factor, USD/kWh
Coal	0.06
Gas	0.07
Oil	0.08
Hydro	0.08
Solar	0.10
Biomass/Biogas	0.08

Table 8
CCS cost factors (european technology platform for zero emission fossil fuel power plants) (Mr Adam Siemenski, 2021),

CCS Cost Component	Cost factor USD/tonne CO_2
CO_2 capture and compression– gas plant	80.23 (• Mr Adam Siemenski, 2021)
CO_2 capture and compression – coal plant	59.51 (• Mr Adam Siemenski, 2021)
CO_2 transport - ship	26.82 (European Technology Platform for Zero Emission Fossil Fuel Power Plants)
CO_2 injection and offshore storage	21.94 (• Mr Adam Siemenski, 2021)

no incentive to decarbonize the energy generation sector. This assessment considers two types of carbon cost i.e. carbon tax and social cost. Carbon tax is a direct form of penalty based on the amount of carbon emitted. Since Malaysia has not implement carbon tax, this study assumes Singapore planned carbon tax rate at USD 50 per tonne CO_2 (Singapore National Environment Agency). This cost will be included as part of the total cost of the scenario This is also the minimum carbon tax that need to be applied to make CCS implementation more palatable to the power generation business.

Social cost of carbon in the other hand is defined as the measure of damage to economic and environment as a result of the emission of additional tonne of carbon. This cost is not monetary in nature, the amount damages are converted into monetary value to simplify its calculation to enable reflection of the impact of continuous emissions. Based on the U.S Department latest estimate, the social cost of carbon is at USD 185 per tonne CO_2 (Rennert et al., 2022). This cost will not be included in the total cost for each scenario, but it will be used as input to consider during the comparative analysis. This applies to all scenarios including Scenario 2 and Scenario 3 where CCS is implemented as there are residual emission post CCS.

2.7.4. Total cost for four scenarios

The total cost of each scenario is the sum of all cost component relevant to the scenario. Electricity generation cost and carbon cost applies to all scenarios, while the CCS cost only applies to Scenario 2 and Scenario 3. Table 9 summarizes the cost assumptions for each scenario. The resultant cost as detailed in Section 4.

2.8. Levelized cost of electricity

Levelized cost of electricity (LCOE) is defined as the measure of the average net present cost of electricity generation for a generator over its lifetime. In this study, the LCOE is calculated between 2023 and 2050. And the discount rate is taken at 4% which is the average discount rate for Malaysia. The LCOE are calculated using equation (4) are as follows. To test the impact of the resultant LCOE are discussed in Section 4.

$$LCOE\left(\frac{USD}{kWh}\right) = \frac{Net\ Present\ Total\ Cost\ (USD)}{Total\ Energy\ Generated\ (kWh)} \tag{4}$$

Table 9
Total cost component - basis.

Cost component	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Electricity generation cost	Case 1 Energy forecast			Case 2 Energy forecast
CCS Cost	Not applicable	Conventional CCS	Solar-supported CCS	Not applicable
Carbon cost	Carbon tax based on residual emission			

2.9. Comparative analysis

2.9.1. Cost-benefit analysis

Cost-benefit analysis was conducted to compare the scenarios based on the ratio of monetary value of benefit over the total cost. Scenario 1 was set to be the baseline case and the analysis is conducted for Scenario 2, 3 and 4. The benefit in this study is represented by the reduction in carbon tax and social cost as the results from reduction in emission. The cost is the summation of generation cost, carbon cost and CCS cost where applicable. A high score indicates that the scenario has significant benefits per unit cost compared to the other scenario. The formula for cost-benefit ratio is given by equation (5). The comparison table is elaborated in Section 4.

$$\frac{Benefit\ (USD)}{Cost\ (USD)} = \frac{Reduction\ in\ Carbon\ Cost\ (USD)}{Generation\ cost + Carbon\ cost + CCS\ Cost\ (USD)} \tag{5}$$

2.9.2. Selection matrix

To corroborate the comparison conducted using cost-benefit analysis, another method was employed which is the selection matrix. A set of selection criteria which are rank based on their importance is used to guide the selection process. A weightage factor is applied to the criteria to represent their significance in the scoring. The five selection criteria include the emission reduction, carbon cost reduction, social cost reduction, net present cost and as well as sustainability. The volume of emission reduction is chosen as the most important criteria to align with the country's net-zero 2050 target. Carbon cost and social cost reduction follows as second and third criteria as they represent the financial benefits and of the interest of investors. Social cost and sustainability are selected as criteria to represent the environmental benefits. Results of the selection matrix is elaborated in Section 4.

2.10. SWOT-PESTEL analysis

To supplement the evaluation of power plant CCS feasibility in Malaysia, qualitative analysis is conducted to identify the barriers and enablers of CCS implementation. Combination of two tools is used in this analysis i.e., the SWOT and PESTEL method. SWOT method identifies the external and internal factors that affects the system by identifying its strength, weakness, opportunities, and threats. PESTEL method on the other hand identifies only external factors, however it does provide a structured framework to analyse a system from political, economic, sociocultural, technological, environment and legal aspect. This study employs a holistic approach by combining SWOT (Strengths, Weaknesses, Opportunities, Threats) and PESTEL (Political, Economic, Social, Technological, Environmental, Legal) analyses to comprehensively identify both external and internal factors that can potentially impact the implementation of CCS in Malaysia's power plants. Specifically, a SWOT analysis is conducted for each element within the PESTEL framework. The highlight of findings is summarized in Table 18.

3. Results and discussion

3.1. Malaysian power plant CCS concept

From the information gathered about existing power plant CCS concept i.e., Petra Nova and Northern Light, evaluation is conducted to ascertain their strengths and weaknesses within the context of Malaysia. This assessment aims to determine their suitability for adoption in Malaysia. Table 10 provides a comprehensive summary of the advantages, disadvantages, and their potential applicability in the Malaysian context for these two CCS designs.

Based on the adaptability assessment above, the recommended carbon capture method for deployment in Malaysia is post-combustion absorption method using amine-based solution. This technology is a mature technology and has been proven successful in other applications.

Table 10
CCS concept advantage & disadvantages.

Design Component	Advantages	Disadvantages	Adaptability in Malaysia
CO_2 Capture Post-combustion absorption	The most established technology for carbon capture in combustion-based process	Requires retrofitting of the capture module into existing process.	Yes. Technology is compatible with Malaysian power plants.
CO_2 Transport Via pipeline	Green solution i.e., no emissions from transport activities. Minimal logistic operations required.	Unscalable-Requires upfront construction of pipeline infrastructure.	No. Full-scale Malaysia-wide CCS will require massive pipeline network that will erode the economic viability of the project.
CO_2 Transport Via CO_2 Tanker Ship	Size of the fleet can be scalable and flexible according to the growing needs.	Additional emissions from shipping operation. Massive logistics operation.	Yes. Malaysia power plants are located along coastal lines.
Carbon Storage in depleted oil & gas reservoir	Sustainable solution as the CO_2 will be permanently stored in the underground formation.	Risk of CO_2 leakage and contamination of ground water.	Yes. There are several candidates of depleted oil and gas fields offshore Terengganu.

It is also compatible with the fossil fuel combustion process and can be retrofitted into current system without the need for major process modification. For CO_2 transport, the recommended solution is transportation via CO_2 tanker ship as it allows for flexibility to develop the full-scale CCS in stages. It is also well-suited for Malaysia as the power plants are located near coastal lines and there is a need to transport carbon from East Malaysia to the storage site in the depleted reservoirs of the Malay basin.

On top of consideration on the existing CCS concept, new alternative solution is also explored to enhance the concept and produce a better fit for Malaysian CCS. Two alternative options that is neither deployed in Petra Nova CCS nor the Northern Light CCS are as listed below. The rationale of considering these alternatives is summarized in Table 11.

-The use of refurbished offshore injection platform and

-The utilization of solar energy to support the additional energy requirement of CCS project i.e., solar-supported CCS.

Table 11
CCS concept – alternatives.

Concept Enhancement	Rationale
Offshore injection platform	-Instead of using sub-sea injection facilities as deployed in Northern Light, the usage of offshore injection platform can significantly reduce cost and complexity of the offshore operation. -It is better suited as Malaysia's depleted oil and gas reservoir are located in shallow water depth which allows for above water operation. -Existing facilities previously used for oil production above the depleted reservoir can be refurbished and utilized to reduce cost to build new infrastructure and reduce waste.
Solar-Supported CCS	-The energy penalty of CCS implementation at power plant are previously supported by burning additional fuel which leads to more emission or by doing nothing, which leads to reduction of the power plant efficiency. -Using solar energy to support the additional energy requirement will eliminate the potential additional emission and maintain the power plant pre-CCS efficiency. -Since Malaysia has good potential for solar energy which can be leveraged without adverse impact to the environment.

Based on the information of proven technologies, best practices, and improvisation for Malaysian context, the concept for full-scale power plant CCS can be established. For capture mechanism, post-combustion capture is the most viable solution for processes that involve fuel combustion like the power generation process in the plant. The most widely used technology is CO_2 adsorption by amine-based solvent that has a capture rate up to 90% CO_2. This CO_2 capture module is to be retrofitted into existing plant or included as part of design for future plants.

In determining the most suitable option for CO_2 transport, a few factors were considered i.e., the location of the power plants, distance from the power plants to the storage site and the need for flexibility of phased development. Ship tankers on the other hand are highly suitable, given that the majority of Malaysia's power plants are situated near the coastline, facilitating easy access for ships. This option also offers flexibility to build the CCS scheme cluster by cluster, thus phasing the investment to be more pragmatic.

Since the identified candidates of CO_2 storage site are located offshore Terengganu, it becomes the logical location for the onshore receiving terminal where CO_2 tanker ships will unload the CO_2 for temporary storage. From the onshore receiving facility, the compressed CO_2 will then be transported via subsea pipeline to an offshore injection platform to be injected into CO_2 storage in an underground depleted formation. Incorporating a solar farm into the concept as an alternative strategy to offset the energy penalty of CCS implementation is a viable option for Malaysia. Besides helping to maintain the amount of energy supplied to the grid, hence preserving the power plant efficiency, using solar energy also fits the country's plan on expansion of renewable resources in the energy mix.

3.2. CCS clustering and infrastructure development strategy

Malaysia's power plants are mostly located along the coastline and fairly distributed in several states as depicted in Fig. 7. Adapting the Northern Light CCS concept, the total of 43 power plants are grouped into twelve clusters based on their location and vicinity to one another. There will be sharing a common facility i.e., compression hub, ship loading jetty and solar farm. This sharing concept will help to optimize the cost of investment. The use of centralized compression hub, CO_2 tanker ships and onshore receiving terminal is chosen as it provides the flexibility to deploy CCS in stages as opposed to transportation via pipeline where an oversized-infrastructure need to be in-place beforehand. By having the onshore receiving facilities, it's also opening the possibility to accept CO_2 from other sources such as industrial CO_2 or even from international sources in the future.

To provide a closer look at the cluster concept, Fig. 8 illustrates the Perak CCS Cluster as an example. CO_2 capture module is to be retrofitted into each of the fossil fuel power plants. The capture module will separate the CO_2 from the flue gas stream to form a concentrated CO_2 stream, and it will be transported via short distance pipeline to a centralized compression facility. This compression facility will be a compression hub that will be shared with other plants within the cluster. From the compression hub, liquefied CO_2 is loaded into CO_2 tanker ship for transport to the onshore receiving facilities which is located onshore Terengganu. From here the liquified CO_2 will be transported via a subsea pipeline to an injection platform to be injected into the depleted reservoir.

On the implementation timeline, it is assumed that only one cluster will be completed each year beginning 2025 as shown in the schedule in Fig. 9. This staged development concept is crucial in a full-scale project as it allows for a better risk management. The first cluster can be considered as a pilot project with minimum capital exposure, where designs are put into test. If there is inadequacy in design or the execution, it will be less expensive to revisit project. This is also why

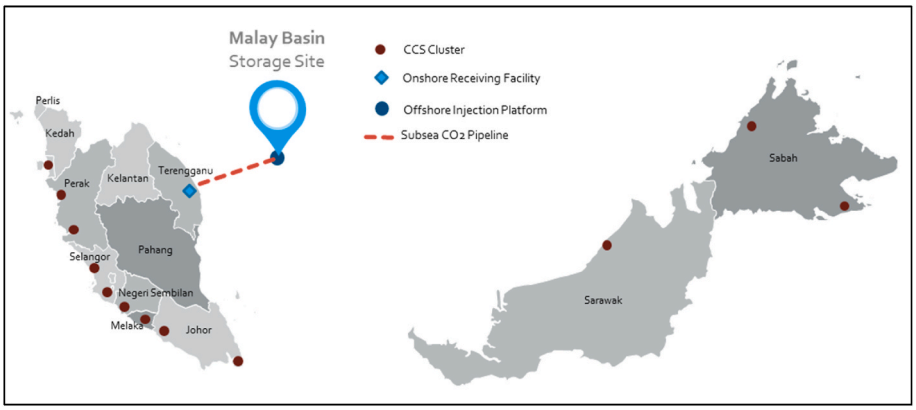


Fig. 7. CCS clusters implementation schedule.

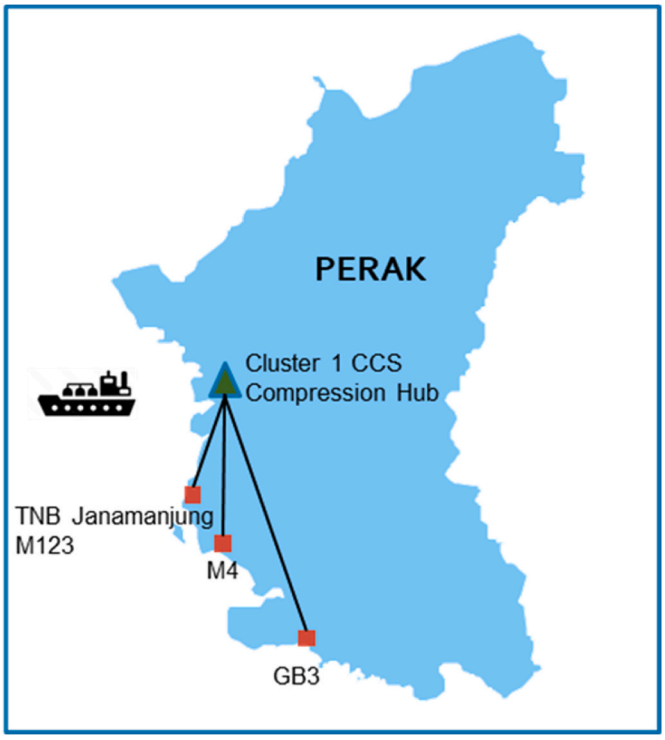


Fig. 8. Perak conventional CCS cluster.

Cluster	Location	Implementation Timeline													
		2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036
1	Perak 1														
2	Neg. Sembilan														
3	Johor 1														
4	Sarawak														
5	Selangor1														
6	Melaka														
7	Johor 2														
8	Penang														
9	Selangor 2														
10	Perak 2														
11	Sabah 1														
12	Sabah 2														

Fig. 9. CCS clusters implementation schedule.

transportation using pipeline may not work in Malaysia as in requires big capital investment upfront.

3.2.1. Conventional CCS and solar-supported CCS

In Scenario 2, the conventional CCS concept is deployed, where the additional energy demands of the CCS module will be met by consuming additional fuel within the plant itself. Consequently, there will be additional emissions as a result of this process. While this may seem counterintuitive to the core goal of CCS, it's important to note that the residual emissions post- CCS implementation are still lower than the total emissions captured. In scenario 3, solar-supported CCS concept is deployed. A solar farm is link to each cluster to supply and offset power to the CCS module. Fig. 10 shows the comparison between the two concepts.

3.2.2. Sizing of solar farm for scenario 3

Solar farm sizing conducted using PVSyst software provides the estimation of land area required to accommodate the number of panels required. For each of the cluster, the solar farm site is chosen based on location, distance from the cluster as well as availability of the land. PVsyst software calculates the area required for a solar farm based on the target electricity output by employing advanced simulation techniques. It considers factors such as solar irradiance, panel efficiency, system losses, and optimization algorithms to estimate the energy production potential of the solar installation. By analysing historical irradiance data, panel characteristics, and system losses, PVsyst determines the optimal layout and configuration of solar panels within the given area to achieve the desired electricity output. The resultant area from the simulation is tabulated in Table 12. Based on the calculation, it is observed that the size of the solar farms is in the mega scale. Accordance to the data obtained from the land office, the ratio between the solar plant to the size of the lands (states) is less than 1%. This is presented in Table 13 indicating the solar farms area and state areas in the respective columns. With the advancement of installation technology and affordable solar panel prices, commencement of large-scale solar plant is feasible under the national energy initiatives and policies.

3.3. Projected energy demand 2050

Fig. 11 shows the historical and projection of Malaysia energy generation Case 1. This scenario is utilized in Scenarios 1, 2, and 3, portraying an extreme situation where Malaysia's energy generation remains heavily reliant on fossil fuels. Consequently, the mix of coal and gas continues to expand until 2050. In contrast, Fig. 12 presents the historical and projected energy generation in Case 2, which is employed in Scenario 4. This scenario represents the extreme opposite of Case 1, assuming that there will be no further growth in fossil fuel generation beyond the current levels. Instead, it envisions a substantial expansion of solar energy capacity to meet the increasing demand. In this outlook, the mix of solar energy is expected to be at 59% of the total energy generation in 2050.

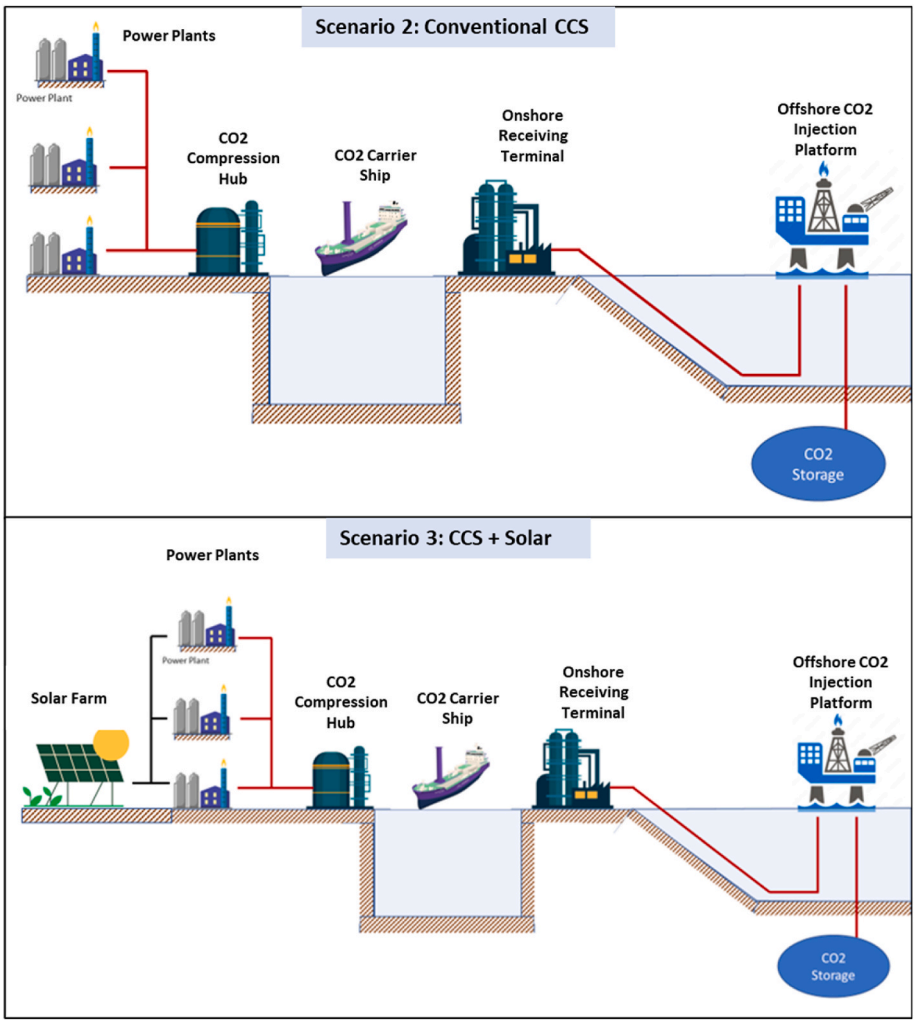


Fig. 10. Malaysia power plants CCS concept.

Table 12
Malaysia power plants CCS concept.

Cluster	Location	Solar Farm Capacity (MW)	Coordinates	Solar Farm Area (km^2)	State Area (km^2)	Solar Farm Relative Size
2	Neg. Sembilan	220	2.636143826468678, 101.80727041419732	5.60	6686	0.04%
3	Johor 1	1830	1.401618799982134, 103.46556816277321	3.53	19166	0.08%
4	Sarawak	1163	2.781284746387287, 112.48285970511226	18.15	124450	0.02%
5	Selangor 1	675	3.49205591228983, 101.29528997628722	30.55	8104	0.01%
6	Melaka	1100	2.357904826099017, 102.07658424743741	3.42	1658	0.38%
7	Johor 2	1232	1.558481116104791, 104.07105106092676	25.59	124450	0.21%
8	Penang	870	5.46426694228589, 100.40970051866404	17.36	1046	0.02%
9	Selangor 2	650	2.5923755014026453, 102.61675534928654	17.79	6686	1.66%
10	Perak 2	623	4.453697515165437, 100.66414449979797	12.87	20973	0.27%
11	Sabah 1	484	4.351237338497506, 118.09954424746925	5.16	73904	0.06%
12	Sabah 2	210	5.6675168019936475, 115.90058943401141	4.94	73904	0.01%

Table 13
Carbon cost for all scenarios.

Resources	Electricity Generation Cost (Bil USD)	
	Scenario 1,2 and 3	Scenario 4
Coal	255.2	155.0
Gas	332.3	171.6
Oil	6.7	5.4
Hydro	84.8	84.8
Solar + Other Renewables	12.4	482.2

3.4. Emission Projection

Fig. 13 shows the resultant emission profiles for all scenarios which reflects the outcome of the energy generation mix and the implementation of conventional CCS or solar supported CCS. Scenario 1 emission profile is expected to have the highest emission volume since there is no CCS implementation. The assumption of Scenario 1 emission profile is that Malaysia will continue to expand its fossil fuel capacity without implementing any CCS.

Scenario 2 emission is lower than Scenario 1 due to implementation

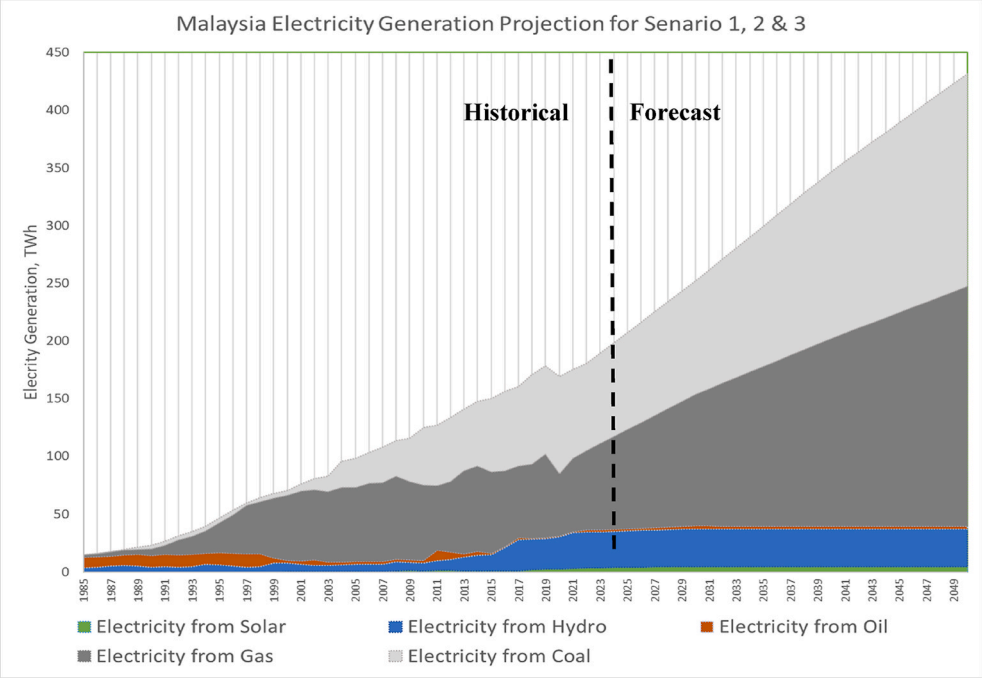


Fig. 11. Case 1 energy generation profile.

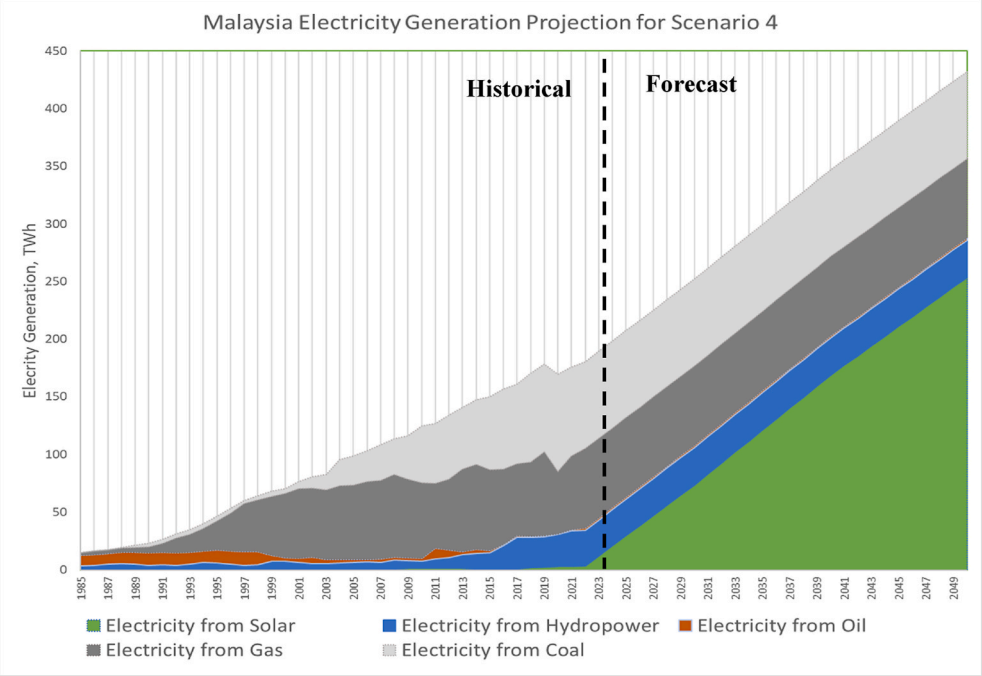


Fig. 12. Case 2 energy generation profile.

of carbon capture, despite the additional emission to support CCS process. The assumption of Scenario 2 profile is that Malaysia will implement CCS to capture the emission from its growing fossil fuel energy generation. There will be additional 20% of fuel burning as a result of CCS energy penalty which adds to the residual emission post CCS. The CCS will be implemented in stages, in which one cluster will come online for each year for a total of 11 CCS clusters. The CCS capture rate is assumed at 90% (Gür, 2022).

Scenario 3 is expected to have the least emission due to implementation of solar-supported CCS. The assumption of Scenario 3 profile is similar to Scenario 2 except for the absence of additional emission as the energy penalty of CCS implementation will be offset by solar energy capacity. Scenario 4 will be the second highest emission as there is no reduction from current emission level. This scenario assumes that Malaysia will prioritize building solar capacity to support the growing demand instead of implementing CCS. It assumes that fossil fuel capacity will not increase further than 2023 level.

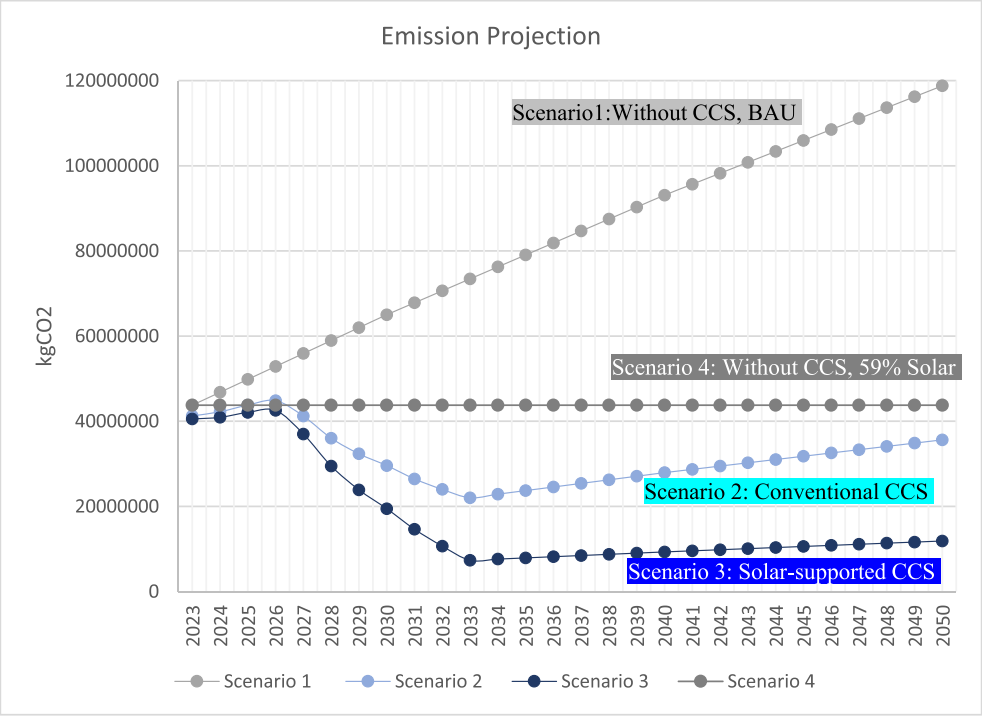


Fig. 13. Emission projection for all scenarios.

3.5. Cost

3.5.1. Electricity generation cost

The electricity generation cost is calculated based on the resources of the energy. Scenario 1, 2 and 3 shares the same energy generation mix therefore cost the same in term of electricity generation. Scenario 4 on the other hand, has higher contribution from renewable energy i.e., solar, hence cost the highest. The electricity generation cost by resource is tabulated in Table 13.

3.5.2. CCS cost

The CCS cost is estimated for Scenario 2 and Scenario 3 where CCS is implemented. The CCS cost factor applied is inclusive of all cost component in the full-chain CCS process i.e., capture cost, compression cost, transfer cost, injection cost and storage cost. Scenario 3 has higher CCS cost due to the additional cost to build and operate solar farm to support CCS energy requirement. This is tabulated in Table 14.

3.5.3. Carbon cost

Carbon cost is estimated in the form of carbon tax to be imposed based on emission level in each scenario where USD 50 is charged per ton CO_2 emitted starting 2023 to 2050 (International Monetary Fund, 2023). This carbon tax rate is similar to the rate implemented in Europe and United States. Scenario 1 has the highest carbon cost due to its higher level of emission as tabulated in Table 15.

3.5.4. Total cost

The total cost in Fig. 14 is the summation of power generation cost, carbon tax cost and CCS cost. For Scenario 2 and Scenario 4, there are only 2 cost components i.e., power generation cast and carbon tax.

Table 14
CCS cost for scenario 2 and scenario 3.

Scenario	CCS Cost (Bil USD)
2	87.8
3	112.8

Table 15
Carbon cost for all scenarios.

Scenario	Carbon Cost (Bil USD)
1	162.48
2	64.96
3	37.10
4	89.94

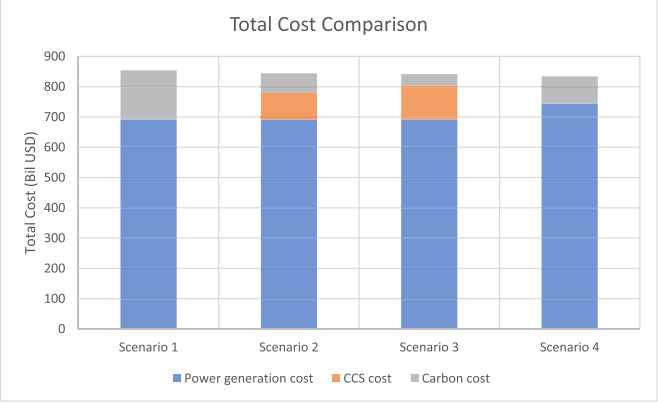


Fig. 14. Emission projection for all scenarios.

While for Scenario 2 and 3, there are a third cost component which is the CCS cost. Generation cost is highest in Scenario 4 due to the higher contribution from the costly renewable energy. Carbon tax is highest in Scenario 1 due to the highest level of emission in this scenario. And CCS cost is higher in Scenario 3 compared to Scenario 2 due to the addition of solar farm to support CCS energy requirement. The comparison of total cost for the scenarios shows that there are comparable to one another. None of the scenario cost significantly more or significantly less than the other. This outcome is due to the inclusion of carbon tax. If carbon tax is removed, Scenario 1 easily becomes the cheapest options. This shows

how significant the impact of carbon tax to facilitate CCS implementation. To identify the best scenario just based on total cost is not advisable in this case.

It is important to note that oil-fired and coal-fired power plants may exhibit variations in their cost structures due to differences in fuel procurement, technology requirements, and operational considerations. However, due to constraints in available data and the inherent complexity of comparing cost factors across different fuel types, a simplified assumption was used in the analysis. This assumption may oversimplify the cost considerations and have potential impact to the accuracy of the findings, hence, further research to explore potential differences in cost factors between oil-fired and coal-fired plants is proposed for future studies. This may include an in-depth examination of fuel procurement costs, infrastructure requirements, and operational expenses associated with each fuel type.

3.5.5. Levelized cost of Electricity Analysis

The levelized cost of electricity (LCOE) analysis is conducted to make comparison in terms of per-unit cost of electricity produced in each of the scenarios. The assessment is conducted up to year 2050, aligned with the timing of net-zero target. The cost components taken into the calculation are the electricity generation cost by resources, carbon cost and CCS cost where applicable. The carbon cost assumption is at USD 50 per ton CO_2 which is at par with the current carbon tax implemented in Europe and United States. The results shows that apart from Scenario 4 that has a higher LCOE due to the more costly per-unit generation for renewable energy component, the value for Scenario 1, 2 and 3 are comparable to one another as shown in Fig. 15. This indicates that when factoring in the carbon tax, the cost of implementing CCS is slightly lower than the cost of emissions without CCS. This finding proves that the LCOE for power generation with CCS is lower than generation without CCS.

To further understand the impact of carbon tax to the LCOE of all the scenarios, the carbon tax is varied between zero to USD150 per ton CO_2 and the resultant LCOE is compared between the scenarios as shown in Fig. 16. The result shows that when carbon tax is equal to USD50 per ton CO_2, the LCOE with or without CCS are almost the same. With the likelihood of increase in the carbon tax rate in the future, CCS scenarios are increasingly becoming more cost-effective and attractive to investors. Conversely, this sensitivity assessment also shows that in the absence of carbon tax i.e. when the carbon tax rate is set at zero, Scenario 1 will emerge as the most cost-effective option, rendering CCS a less attractive option. Hence, it is crucial that carbon tax is imposed to emitting indutries to create the financial incentives to invest in CCS.

3.6. Comparative analysis

3.6.1. Cost-benefit comparison

The first method of comparative analysis is the cost-benefit analysis

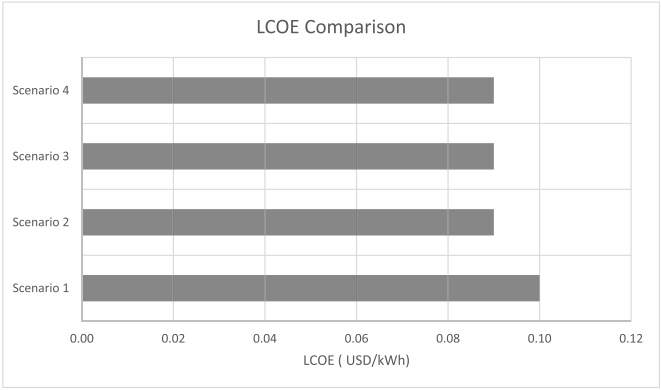


Fig. 15. LCOE for all scenarios.

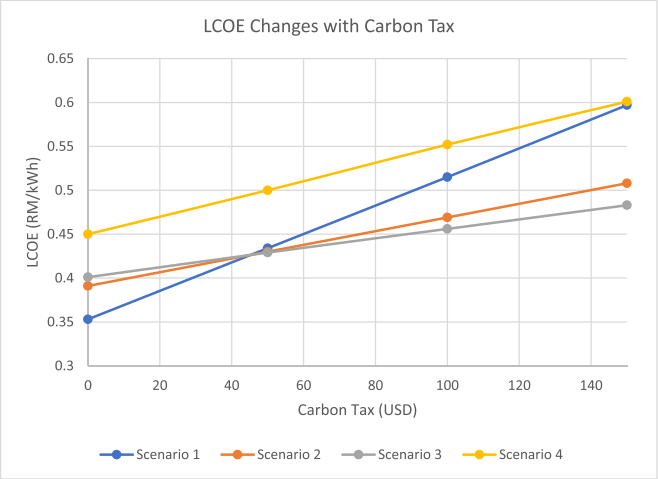


Fig. 16. LCOE changes with carbon tax.

Table 16
Cost-benefit analysis.

Cost Benefit	Scenario 1 (baseline)	Scenario 2	Scenario 3	Scenario 4
Emission (Gigaton CO_2)	3.25	1.30	0.74	1.80
Emission Reduction (Gigaton CO_2)	0	1.95	2.51	1.45
Emission Reduction (%)	0.00%	60.02%	77.17%	44.64%
Carbon Tax (Bil USD)	162.48	64.96	37.10	89.94
Carbon Tax Reduction (Bil USD)	0	97.52	125.38	72.54
Total Cost (Bil USD)	854.04	844.29	841.51	989.00
Benefit/Cost ratio	–	**0.12**	**0.15**	**0.07**

as shown in Table 16. Scenario 1 serves as the baseline against which the other scenarios are compared to. The assessed benefit in this analysis is the reduction of emissions from the baseline level in Scenario 1, which is quantified in monetary terms through the decrease in carbon tax payments as the result of lower emissions. The carbon tax reduction is divided with the total cost to give the benefit over cost ratio of each scenario. The analysis results shows that Scenario 3 has the highest ratio followed by Scenario 2 and then Scenario 4. This suggest that implementing solar-supported CCS is the best option as it gives highest benefit per every dollar invested. Implementation of conventional CCS also yields a competitive ratio, further supporting the viability of CCS adoption. On the other hand, Scenario 4 gives lowest ratio because of the lesser amount of emission reduction as well as the higher total cost.

3.6.2. Selection matrix

The second method of comparative analysis is the selection matrix. The selection is conducted based on the five criteria used to weigh the options i.e., emission reduction, carbon tax reduction, social cost reduction, total cost and sustainability as shown in Table 17. In terms of emission and carbo tax reduction, Scenario 3 has the highest score followed by Scenario 2, 4 and 1 as projected in Fig. 11. The social cost refers to the damages to the environment as a result of emission. Options with the highest reduction of social cost which is Scenario 3 and Scenario 2 gets higher score as these two options emits the least amount of carbon. Total cost refers to the summation of energy generation cost and carbon tax cost and including the CCS cost for scenario 2 and 3. Options with lowest total cost i.e., Scenario 3 and 2 gets higher score. In terms of sustainability, option with more contribution from renewable resources gets higher score as compared to options with higher mix of fossil fuel resources. The result of the selection matrix shows that Scenario 3 has

Table 17
Selection matrix between scenarios.

		Scenario 1 - Business as usual		Scenario 2 – Conventional CCS		Scenario 3 – Solar supported CCS		Scenario 4–56% RE	
Criteria	Weightage	Score	Total	Score	Total	Score	Total	Score	Total
Emission Reduction	5	1	5	3	15	4	20	2	10
CT Reduction	4	1	4	3	12	4	16	2	8
Social cost reduction	3	1	3	3	9	4	12	2	6
Total Cost	2	1	2	2	4	3	6	4	8
Sustainability	1	1	1	2	2	3	3	4	4
			15		42		**57**		36

the highest total score followed by Scenario 2. This means implementing CCS is better than relying solely on the expansion of renewable energy to reduce the level of emission. Despite the aggressive growth of renewable energy assumed in Scenario 4, there are still high dependency on fossil fuel in the Malaysia energy landscape. Hence, the level of emission in Scenario 4 is higher than Scenario 2 and 3.

3.6.3. Benchmarking with existing power plant CCS

Results from the comparative analysis suggest that solar-supported CCS, implemented by clustered and developed in stages is the best scheme to implement to decarbonize Malaysia energy sector. Comparing this with the existing power plant CCS such as Petra Nova and Boundary Dam CCS, there are a few distinct features that set this plan apart from its predecessor, with the aims to improve the likelihood of success based on the learning on previous experiences.

This proposed CCS scheme is not impacted by the volatility of oil prices, a key factor that contributed to the challenges faced by Petra Nova CCS. This is due to the improved economic model wherein carbon is not treated as a tradable commodity but rather as stored permanently.

Table 18
CCS barriers and enablers.

	Barriers	Enablers
Technology	Capture technology still at low efficiency.	Capture technology must be close to 90% efficiency to reap benefit of low carbon economy. More R&D required.
Political	There is no local expertise/ experience to develop regulatory framework for CCS implementation in the country.	International agreement for resources sharing and regional collaboration.
Economics	- There is currently no economics model/market for CCS project in Malaysia. -Malaysia currently not implementing carbon pricing which can generate revenue for CCS. -There is no carbon utilization market in the manufacturing industry which can create revenue for CCS. -There are no financing mechanism in-place except for funding from government.	-Introduction of carbon pricing and carbon trading to create revenue stream for CCS. -Private sectors funding opportunities especially from industry heavily impacted by CCS can be secured for R&D, infrastructure building, etc.
Social	Low public acceptance on the mechanism for CO_2 storage which is deem unsafe.	Increase public engagement and built low carbon technology in syllabus.
Environmental	Carbon capture process are energy intensive which require to burn more fossil fuel and releases more emission.	Built solar supported CCS.
Legal	The liability for monitoring CCS storage site can be a dispute among the parties involved and often become a showstopper.	Government to consider assuming the responsibility for monitoring and surveillance of the storage site as owner of mineral and organic resources.

Table 19
SWOT-PESTEL results Part 1.

Implementation of widescale CCS to decarbonize the energy sector in Malaysia		
Political	Strength The government of Malaysia had pledged a net-zero target by 2050. The state government in Sarawak has announced plans to regulate the CCS activities in the state.	Weakness There is no local expertise/ experience to develop regulatory framework for CCS implementation in the country.
	Opportunity international agreement for resources sharing regional collaboration	Threats Securing budget for CCS can be a political issue as it requires huge funding and also competes with other national priorities.
Economics	Strength Government willingness to subsidize/fund the studies of CCS implementation in Malaysia.	Weakness There is currently no economics model/market for CCS project in Malaysia. Malaysia currently not implementing carbon pricing which can generate revenue for CCS. There is no carbon utilization market in the manufacturing industry which can create revenue for CCS. There are no financing mechanism in-place except for funding from government.
	Opportunity Introduction of carbon pricing and carbon trading to create revenue stream for CCS. Private sectors funding opportunities especially from industry heavily impacted by CCS can be secured for R&D, infrastructure building, etc.	Threats With no proven success in the region, financial institution does not have the incentives to fund CCS projects.
Social	Strength Malaysian publics are increasingly becoming aware of the impact of emission to climate change	Weakness Low public acceptance on the mechanism for CO2 storage which is deem unsafe.
	Opportunity There are several avenues for public engagement that can be utilized to increase awareness and to garner public support.	Threats Difficult to garner support as CCS might be perceived by the public as the government's scheme to give 'license' to continue fossil-fuel-power-generation with no genuine interest to decarbonize.

And the implementation of carbon taxes serves as a robust financial incentive, further reducing the reliance on oil prices. In terms of scalability, the staged development plan enabled through the shipping transportation mode, will provide flexibility to optimize as needed.

3.6.4. SWOT-PESTEL findings

Tables 18–20 summarize the key findings of SWOT-PESTEL. The key barriers identified for implementation of power plant CCS in Malaysia

Table 20
SWOT-PESTEL results Part 2.

Implementation of widescale CCS to decarbonize the energy sector in Malaysia		
Technology	Strength The technology for power plant CCS has been proven in operation i.e., Petra Nova and Boundary Dam CCS.	Weakness There is no local expertise/ experience to build and operate CCS system in Malaysia.
	Opportunity Advancement of CCS technology through R&D activities can lead to optimization of the performance as well as cost, improving CCS economics viability.	Threats Competing with advancement of renewable energy technologies. The RE might become very competitive that CCS becomes obsolete/ unjustifiable.
Environmental	Strength CCS can potentially capture 90% of emission from being release into the atmosphere, hence slowing global warming.	Weakness Carbon capture processes are energy intensive where additional energy is required to run the system. This is usually achieved by burning more fossil fuel which releases more emission. Or this can be achieved by using RE i.e., solar which also have carbon footprint from land clearing and solar farm construction activities.
	Opportunity Malaysia has several depleted oil and gas reservoir which are good candidates for storage sites. These reservoirs are located offshore, away from population or wildlife.	Threats Potential leakage of carbon from the storage site through subsurface fractures or compromised seal. Some carbon capture technology has adverse environmental impact in terms of intensive water consumption and wastewater disposal. Construction activities of massive CCS infrastructure will have negative environmental impact i.e., change of land use, destruction of biodiversity.
Legal	Strength Malaysia has international pledge to reduce its emission as the participating member of the Paris Agreement 2016. This provides the motivation for CCS.	Weakness No existing regulatory framework to control and manage CCS scheme. It is unclear which party will assume the responsibility of monitoring and managing the carbon storage site. Since this a huge liability it is often avoided.
	Opportunity	Threats Although Malaysia ratified the Paris Agreement, as a non-Annex I country, it has no obligation to reduce emission. Wide-scale CCS might involve cross-border operation e.g., straddling or connected storage site, transportation by ship, which can be a source of legal disputes and concerns.

are from the economics aspect, which is due to the absence of carbon market (Haniffa et al., 2022). From the political aspects, it is due to the absence of policy and regulations that will facilitate CCS implementation. Another potential barrier is from the social aspect where there could be potential protest from the public with regards on CO_2 injection. However, this risk is low as the plan is to inject the CO_2 at a location offshore, away from residential area.

4. Conclusion and recommendation

This paper presented significant findings on the feasibility of CCS implementation in Malaysia's fossil fuel power plants. This work analysed the benefit and cost of implementation of CCS supported by comparative studies of various energy scenarios. This provides a holistic view in deciding whether CCS is a viable solution for decarbonization. In addition to that, qualitative assessments were also conducted to identify the barriers and subsequently the key enablers of CCS implementation in Malaysia's fossil fuel power plants. The following are the key findings of the assessment.

The potential carbon emission reduction that can be achieved through a nation-wide CCS implementation at fossil fuel power plants are up to 60 % in Scenario 2. This can be further improved up to 77 % emission reduction by supporting the CCS process with solar energy as in Scenario 3. When carbon cost is considered, the total cost of conventional CCS implementation as well as solar -supported CCS implementation is comparable with the cost of energy generation without CCS. And this is also reflected in the LCOE comparison of Scenario 1, 2 and 3 where there is only little variance between the three. Transporting the liquified CO_2 by ship allows for a staggered expansion of the CCS scheme. It provides the flexibility as well as risk mitigation plan required in billion-ringgit scale investment. Based on the techno-commercial assessment, it can be concluded that a wide scale power plants CCS can successfully be implemented if the following conditions can be met.

- Carbon tax is to be imposed to emitting industry to create the motivation for carbon capture and storage. Without carbon tax, there will be no financial incentives to decarbonize the energy sector and to implement CCS. This study shows a minimum of USD 50 per tonne CO2 carbon tax is required to make CCS economically viable.
- The carbon capture rate at the plants is reaching the theoretical capture rate of 90 %. This efficiency is required to reap the benefits of low carbon economy. If the capture rate is low, residual emission will increase as well as carbon tax, making the project more economically disadvantaged. Historically, the actual rate for post combustion capture is within the range of 60–85%.
- The implementation is to be conducted in multiple smaller scale stages to reflect a more realistic investment commitment as well as allowing room for optimization as a result of the learning curve.
- The power plants are clustered together to allow for infrastructure sharing and leverage on of the economics of scale.
- The social cost of carbon emission is to be seriously taken into consideration in the cost benefit analysis to reflect the true cost of emission to the environment. Implementation of carbon tax alone will not be sufficient to replace the damages caused by climate change.

In terms of advancement of the knowledge and technology of CCS, it is recommended to conduct research aiming to reduce the energy penalty from its implementation and improve its efficiency. It is recommended that a detail analysis to be conducted on the following subject to substantiate the findings and conclusions.

- To assess the feasibility of CCS when capture rate at the power plants are lower than theoretical value of 90%.
- To evaluate the technical viability of retrofitting carbon capture module into existing power plants especially decades old plants.
- To evaluate the techno-commercial feasibility of mega solar-farm projects supporting the CCS scheme.
- To consider the energy mix scenario in the latest national plan to reflect a more accurate emission scenario.

Malaysia is still falling short in terms of readiness to implement CCS due to absence of legal framework to regulate the operation of CCS, manage its liability and minimize its exposure. This study findings and

conclusion can provide information for various stakeholders. For example, Ministry of Natural Resources, Environment and Climate Change can formulate the policies and regulations to facilitate decarbonization through CCS technologies. Second, Tenaga National Berhad and Independent Power Plant owners can develop better understanding on the technical challenges and the opportunities of CCS implementation in their operation. Third, Ministry of Finance, investors and financial institution can explore the feasible financing scheme to fund CCS projects.

The implications of this study are multifaceted and underscore the pivotal role of policy, technology, and finance in facilitating the implementation of CCS technology in Malaysia's fossil fuel power plants. Policymakers should consider introducing carbon pricing mechanisms to incentivize industry adoption of CCS, while concurrently developing a robust legal framework to regulate operations and manage associated liabilities. Furthermore, advancing CCS technology through research and development initiatives is imperative to enhance efficiency and reduce the energy penalty, thereby making large-scale CCS schemes economically viable. Financial institutions and investors should explore funding opportunities for CCS projects, leveraging private sector investment alongside government subsidies and carbon pricing mechanisms. Moving forward, the study suggests conducting detailed evaluations of CCS feasibility scenarios, engaging stakeholders through capacity-building initiatives, and collaborating internationally to accelerate Malaysia's transition to a low-carbon economy. By embracing these recommendations, Malaysia can play a pivotal role in global efforts to mitigate climate change while contributing to the achievement of UN-SDG and net-zero targets by 2050.

In comparison to the studies conducted in various countries such as the United States, Norway, Canada, Australia, and the United Kingdom, this study on assessing the feasibility of CCS implementation in Malaysia's fossil fuel power plants offers unique insights tailored to the Malaysian context. While these international studies provide valuable data and perspectives on CCS technology, our research focuses specifically on Malaysia's energy landscape, analysing the potential carbon emission reductions achievable through different scenarios of CCS implementation, including solar-supported CCS. Furthermore, our study addresses the economic viability of CCS deployment by comparing the total costs of conventional and solar-supported CCS with the costs of energy generation without CCS, highlighting the potential for comparable costs and minimal variance in levelized cost of electricity (LCOE). Additionally, the findings emphasize key conditions necessary for successful CCS implementation in Malaysia, including the importance of carbon taxation, achieving high capture rates, staged implementation, clustering of power plants, and consideration of social costs in cost-benefit analyses. Moreover, this study underscores the need for policy development, technological advancements, and financial support to facilitate CCS deployment in Malaysia, outlining recommendations for stakeholders such as government agencies, power plant operators, and financial institutions to drive decarbonization efforts through CCS technology adoption. Overall, our research provides a comprehensive analysis of the challenges and opportunities associated with CCS implementation in Malaysia, serving as a valuable resource for policymakers, industry stakeholders, and investors seeking to accelerate the transition to a low-carbon economy and mitigate climate change impacts.

List of Abbreviations

BCG	Boston Consulting Group
CAPEX	Capital Expenditure
CCS	Carbon Capture and Storage
CCUS	Carbon Capture, Utilization, and Storage
CO2	Carbon Dioxide
COP	Conference of the Parties
COP15	Conference of the Parties 15
EOR	Enhanced Oil Recovery
EOR	Enhanced Oil Recovery
FiT	Feed-in Tariff
GDP	Gross Domestic Product
GHG	Greenhouse Gas
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
LCOE	Levelized Cost of Electricity
LSS	Large Scale Solar PV
m	Meters
MoU	Memorandum of Understanding
MtCO2eq	Metric Tons of Carbon Dioxide Equivalent
Mtpa	Million Tons Per Annum
MW	Megawatt

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Ernieza Nasir: Writing – original draft, Methodology, Formal analysis. **Yun Ii Go:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Not applicable.

NDC Nationally Determined Contribution
NEM Net Energy Metering
PESTLE Political, Economic, Social, Technological, Legal, Environmental
PETRONAS Petroliam Nasional Berhad
PETRONAS Petroliam Nasional Berhad (Malaysia's national oil and gas company)
PVSyst Photovoltaic System Simulation Software
RE Renewable Energy
RM Malaysian Ringgit
RMK-12 12th Malaysian Plan
SWOT Strengths, Weaknesses, Opportunities, Threats
TNB Tenaga Nasional Berhad (Malaysia's energy utility company)
UN United Nations
USD United States Dollar
WWF World Wide Fund for Nature

Appendix. Power Plants Data [Source: Grid System Operator, Malaysia]

Location	Name	Fuel	Capacity (MW)
Johor	Tanjung Bin Power Sdn Bhd	Coal	2100
Johor	Tanjung Bin Energy Sdn Bhd	Coal	1000
Johor	TNB Pasir Gudang Energy Sdn Bhd	Gas	275
Johor	Pengerang Power Sdn Bhd	Gas	600
Johor	Southern Power Generation Sdn. Bhd. (SPGP)	Gas	1440
Melaka	Pahlawan Power Sdn Bhd	Gas	322
Melaka	Panglima Power Sdn Bhd	Gas	720
Melaka	Edra Melaka Power Plant (Block 1, 2, 3)	Gas	2242
Negeri Sembilan	Kapar Energy Ventures Sdn Bhd	Coal	1474
Negeri Sembilan	Jimah Energy Ventures Sdn Bhd	Coal	1400
Negeri Sembilan	Jimah East Power Sdn Bhd	Coal	2000
Negeri Sembilan	S.J Tuanku Jaafar, Port Dickson	Gas	703
Negeri Sembilan	S.J Tuanku Jaafar, Port Dickson	Gas	708
Penang	Prai Power Sdn Bhd (Malakoff)	Gas	350
Penang	S.J Gelugor	Gas	310
Penang	TNB Prai Sdn Bhd	Gas	1071.43
Perak	TNB Janamanjung Sdn Bhd (M123)	Coal	2070
Perak	TNB Janamanjung Sdn Bhd (M4)	Coal	1010
Perak	TNB Manjung Five Sdn Bhd	Coal	1000
Perak	GB3 Sdn Bhd	Gas	640
Perlis	Teknologi Tenaga Perlis Consortium Sdn Bhd	Gas	650
Putrajaya	S.J Putrajaya	Gas	249
Sabah	SJ Patau Patau	Gas	103.4
Sabah	SPR	Gas	100
Sabah	Kimanis	Gas	285
Sabah	Teluk Salut	Gas	190
Sabah	Rugading	Gas	190
Sabah	Sepanggar	Gas	100
Sabah	Malana	Oil	18
Sabah	SJ Sandakan	Oil	36
Sabah	Batu Sapi	Oil	17.4
Sabah	SJ Tawau	Oil	32.7
Sarawak	Balingian Power Plant	Coal	600
Sarawak	Mukah Power Station	Coal	270
Sarawak	Sejingkat Power Plant	Coal	200
Sarawak	PPLS Power Generation Plant	Coal	110
Sarawak	Sarawak Power Generation Plant	Gas	515
Sarawak	Pujut Power Station	Gas	104
Selangor	Kuala Langat Power Plant Sdn. Bhd. (KLPP)	Gas	675
Selangor	Kapar Energy Ventures Sdn Bhd	Gas	578
Selangor	Segari Energy Ventures Sdn Bhd	Gas	1303
Selangor	TNB Connaught Bridge Sdn Bhd	Gas	375
Terengganu	YTL Power Generation Sdn Bhd (Paka)	Gas	585

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